

Mining Spectral Intervals for Backdoor Vulnerability Analysis in Hyperspectral Image

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Abstract—The ordered spectrum of a hyperspectral image (HSI) makes trigger support a distinct design choice in backdoor poisoning. We propose DSI-LAMM, a training-time attack using target-conditioned spectral intervals to study compact spectral backdoors. Target-background discriminability, block-wise consistency, and redundancy control select short intervals under a band budget. A bounded directional shift acts on this support; prototype margins and post-shift target proximity rank carriers, with source-class quotas and class/spatial caps governing selection. On four public benchmarks, 3% dirty-label poisoning yields dataset-averaged triggered target-class rates of 92.2–99.4%, while clean overall accuracy remains within 1.11 percentage points of the clean reference. Relative to their all-band versions, Blended and WABA on the mined supports improve target control and reduce patch-level mean absolute perturbation by approximately 84–90%. The supports contain both target-dependent and shared regions, with repeatability varying across scenes. Matched-carrier comparisons on Pavia and Houston2013 identify several effective compact layouts. Clean-reference tests show large additional trigger responses after poisoning. These findings motivate training-data audits that examine localized spectral changes together with the source-class composition of contributed samples.

Index Terms—Data Poisoning, Backdoor Attack, Hyperspectral Image, Adversarial Pattern Discovery, Spectral Motif Mining

I. INTRODUCTION

Hyperspectral image (HSI) classification derives land-cover information from densely sampled spectral measurements and their spatial context. The ability to distinguish visually similar materials makes it useful for crop identification, environmental monitoring, and urban mapping [1]. Deep spectral-spatial models have strengthened this capability by learning joint representations of local texture and spectral signatures, from hybrid convolutional networks to spectral transformers [2], [3]. Their reliability also depends on the conditions under which these representations are learned and used. Studies of adversarial examples and HSI defenses have shown that small changes to an input can disrupt predictions even when clean performance is high [4], [5]. A separate risk arises before inference, when the model learns from contaminated training data. Backdoor poisoning uses this opportunity to associate a recurring perturbation with a target label, allowing normal behavior on clean inputs to coexist with trigger-induced errors [9], [10]. For an HSI training pipeline that accepts

contributed samples and annotations without independently checking both, such an association can be introduced through a small part of the training set. Understanding this risk requires examining the spectral changes from which the association is learned, rather than relying on clean accuracy alone.

The ordered and redundant spectrum of HSI gives this training-time risk a structure that is not captured by treating every band as an interchangeable image channel. Neighboring bands measure nearby wavelengths, and information useful for distinguishing land-cover classes can extend across several spectral regions [1], [6]. This structure has long motivated band selection for classification, but it also provides a basis for constructing perturbations that leave most of the spectrum unchanged. Existing backdoors offer several ways to alter an image, including spatial patches, blending, sinusoidal signals, and geometric warping [9]–[12]. Remote sensing methods further use wavelet coefficients or selected channels to embed triggers [13], [14]. The spectral support of a trigger, meaning the bands on which it acts, nevertheless remains distinct from the rule used to modify their values. Applying a trigger across all bands entangles its success with a broad spectral perturbation; reporting a single mean distortion then obscures where that perturbation is concentrated. Making the support an explicit part of the construction allows localized changes to be related to target-class evidence and compared across trigger forms. This connection is important for understanding how a corrupted classifier can retain its normal task while responding to changes confined to a few spectral regions.

Our starting point is to use target-background spectral contrast to construct these localized changes from the training data. A small collection of intervals retains neighboring measurements while allowing separated regions of the spectrum to contribute to the same target association. Selecting such intervals involves more than taking the largest individual band scores: adjacent scores can describe redundant evidence, and a strong contrast in one spatial region may not recur elsewhere. We combine spectral redundancy and spatial variation as criteria for constructing short target-conditioned supports. The same data dependence also affects the samples that carry the perturbation. A bounded shift has different effects on spectra from different source classes, and a few highly ranked samples can leave much of the non-target population unrepresented. A

limited poisoning budget links these two choices: the available carriers determine how widely the selected spectral pattern is represented in the contaminated data. We use class statistics to describe the spectral direction of the trigger and changes in source-to-target prototype relationships to assess candidate carriers, with explicit allocation across source classes. This couples the spectral pattern to the training examples through which the target association is introduced.

We develop DSI-LAMM, a directional spectral-interval motif mining attack, to implement this construction and study compact HSI backdoors. In the dirty-label setting considered here, the adversary can inspect training spectra, labels, and spatial locations and can modify and relabel a limited subset before victim training. DSI-LAMM combines target-background discriminability with block-wise consistency, smooths the resulting band scores, and selects short intervals while limiting redundancy. It defines a bounded directional motif on this support and ranks carriers by prototype-margin change and post-shift target proximity, with source-class quotas and spatial allocation controls. The construction requires no victim-model queries, and the classifier continues to receive the full spectrum. The selected intervals remain an explicit output, which can be compared across targets and training splits or reused with other trigger rules.

Our contributions are as follows.

- We formulate compact HSI backdoor construction through target-conditioned spectral intervals, separating the spectral support of the trigger from its injection rule and from the full-spectrum model input.
- We develop DSI-LAMM, which combines interval-level spectral selection, a bounded directional motif, and prototype-margin carrier ranking with source-class quotas and spatial allocation controls.
- We evaluate DSI-LAMM on four HSI benchmarks and examine support reuse, layout repeatability, and victim-model sensitivity. Matched-carrier and clean-reference controls assess support location and carrier selection.

II. RELATED WORK

A. HSI Classification and Band Selection

HSI representation learning combines spectral signatures with spatial context. Early deep belief networks and spectral convolutional networks learned representations from hyperspectral data [28], [29], followed by architectures that fuse multilevel features [30]. HybridSN combines a 3-D convolutional hierarchy with subsequent 2-D spatial processing [2]. SpectralFormer groups neighboring bands into local spectral embeddings, while spectral-spatial feature tokenization represents local information as tokens for a transformer [3], [31]. These methods adapt sequence and image representation learning [32] to the correlations within an HSI cube.

Band selection reduces input redundancy by retaining informative channels [6]. The mRMR criterion balances label relevance against dependence among selected features using mutual information [7]. The optimal clustering framework for

HSI groups bands and selects representatives, using inter-band structure to avoid repeatedly choosing similar measurements [8]. These approaches seek a reduced representation for classification. We instead select a perturbation support for one target class while leaving the model input unchanged. Fisher discriminability and correlation serve as established statistical ingredients; interval constraints organize them into a compact support on which a bounded spectral trigger is defined.

B. Backdoor Attacks and Defenses

Backdoor attacks establish a trigger-target association during training [9], [10], with different assumptions about label manipulation and trigger placement. BadNets uses a local pattern, Blended attacks mix a pattern with the image, and WaNet uses spatial deformation [9], [10], [12]. SIG introduces an additive sinusoidal signal in a setting without label poisoning [11]. BackdoorBench documents how trigger design and training protocol affect evaluation [38]. Our HSI adaptations use the common dirty-label protocol and entrywise perturbation bound specified in Sec. V.

Defenses inspect different consequences of the learned association. Spectral Signatures examines singular directions in training-feature matrices, whereas Activation Clustering searches for separated groups of within-class activations [15], [21]. Here, spectral analysis concerns a feature matrix, not HSI wavelengths. Neural Cleanse compares the sizes of reconstructed triggers across labels [22]. STRIP measures predictive behavior under input mixing, and IBD-PSC examines consistency under parameter scaling [26], [27]. These principles distinguish feature-space separation, trigger reconstruction, and response consistency; the number of altered input bands alone does not determine their detection outcomes.

C. Remote Sensing Robustness and Backdoors

Remote sensing robustness studies connect adversarial behavior to learned representations. Xu and Ghamisi generate transferable adversarial examples by attacking shallow surrogate features for scene classification and semantic segmentation [4]. For HSI, S³ANet combines multiscale spatial attention with a global spectral transformer to defend against adversarial input perturbations [5]. These studies examine susceptibility to changes at inference time; backdoor poisoning changes the training examples through which the decision function is learned.

WABA embeds triggers by blending low-frequency wavelet coefficients and retaining the original high-frequency detail coefficients [13]. These spatial-frequency components differ from intervals along the HSI wavelength axis. The sub-band attack of Islam *et al.* uses explainability-guided channel selection [14], establishing channel choice as part of remote sensing backdoor design. Our construction selects short, target-conditioned intervals from labeled training spectra and applies a bounded directional shift on their union. Keeping this support separate from the injection rule permits its reuse by different triggers. The resulting comparison concerns spectral

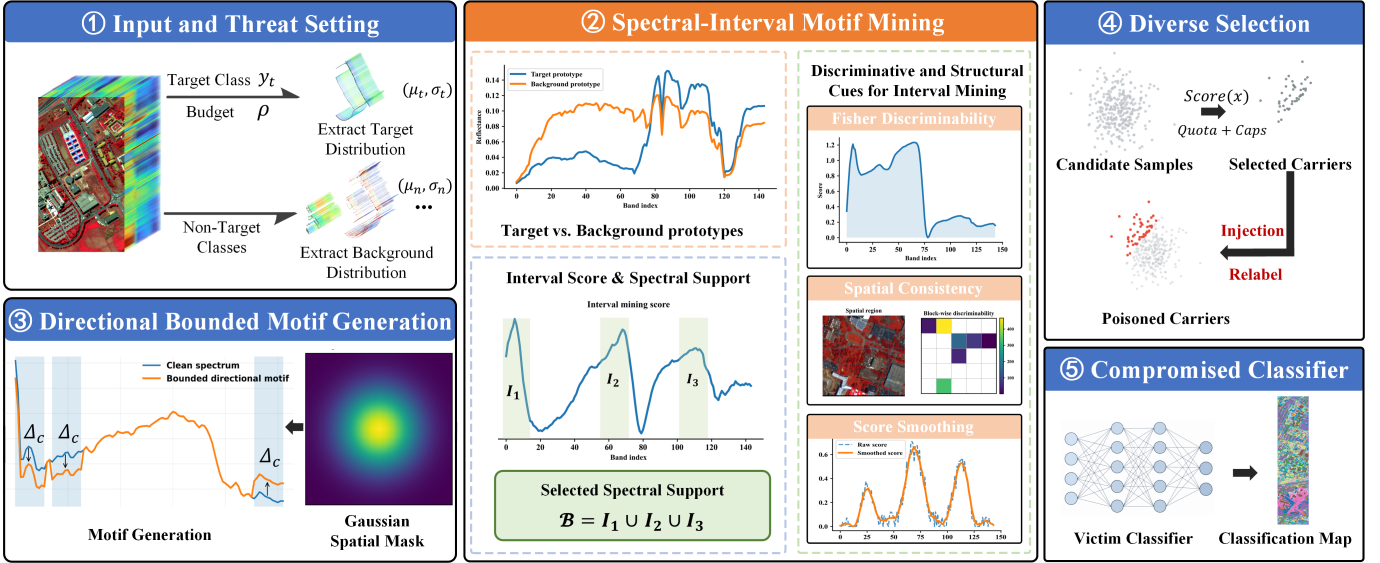


Fig. 1. Schematic workflow of DSI-LAMM. (1) Training samples, a target class y_t , and a poisoning rate ρ define the input. (2) Target-background discriminability, spatial consistency, and score smoothing guide spectral-interval selection with redundancy control. (3) A Gaussian spatial mask distributes a bounded directional shift over the selected bands. (4) Prototype-margin scoring, source-class quotas, and class/spatial caps guide carrier selection; the selected samples are then modified and relabeled. (5) The victim is trained on the full resulting training set, including the poisoned carriers. The classification map illustrates the classifier output.

support within a scene, rather than transfer of an adversarial perturbation between victim networks.

D. Spectral Motifs and Budgeted Selection

Time-series motif discovery searches for recurring local subsequences [16], [34], and the matrix profile represents nearest-neighbor subsequence distances for motif and discord search [17]. Shapelet learning uses local subsequences to distinguish classes [33]. HSI spectra admit a related interval representation, but our motif is constructed from target-background statistics rather than retrieved as a repeated subsequence. Labels determine the interval scores and shift direction, allowing the selected regions to be compared across targets and training splits.

Coresets approximate a data-dependent objective with a small, sometimes weighted subset [18], [35]; gradient-based selection can approximate training updates [19]. Other approaches use early training scores, geometric coverage, or lower-cost proxy models to select informative examples [20], [36], [37]. Here, selected samples carry the spectral motif and return to the full training set after modification. Prototype-margin ranking describes their response to the shift, while quotas regulate source-class representation. The objective is to establish a target association through a limited carrier set, rather than to approximate clean training on a smaller dataset.

III. PROBLEM FORMULATION

We study HSI backdoor vulnerability through a training-time poisoning attack with compact spectral support.

A. Notation

Let $\mathcal{D} = \{(x_i, y_i, r_i)\}_{i=1}^N$ denote an HSI classification dataset. Each normalized sample $x_i \in [0, 1]^{P \times P \times C}$ is a

spatial-spectral patch centered at a labeled pixel, where $P \times P$ is the spatial window size and C is the number of bands indexed in spectral order. The label $y_i \in \{0, 1, \dots, K-1\}$ denotes the land-cover class, and r_i gives the spatial location of the central pixel. We use $s_i \in [0, 1]^C$ for the central spectral vector. The training, validation, and test partitions are fixed before poisoning and are denoted by $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{val}}$, and $\mathcal{D}_{\text{test}}$, respectively. A classifier $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ maps each patch to a class label.

B. Threat Model

We assume the adversary can inspect training patches, labels, and spatial locations, and alter both the inputs and labels of a budgeted subset before training. This setting represents a training-data pipeline that accepts externally supplied samples with editable labels. We construct the trigger from training-set spectral statistics, without access to the victim architecture, parameters, or predictions. Validation and test samples are excluded from trigger construction and carrier selection.

The attacker specifies a target class y_t and a nominal poisoning rate $\rho \in (0, 1)$. Let B denote the corresponding integer budget in training samples. The candidate pool contains the non-target training samples:

$$\mathcal{C}_t = \{(x_i, y_i, r_i) \in \mathcal{D}_{\text{train}} : y_i \neq y_t\}. \quad (1)$$

The attacker selects $\mathcal{D}_{\text{sub}} \subseteq \mathcal{C}_t$ with $|\mathcal{D}_{\text{sub}}| \leq \min(B, |\mathcal{C}_t|)$ and applies a trigger injection function $\mathcal{T}_\phi$. Each selected patch is modified and relabeled as y_t . Poisoning is defined at the training-sample level, giving

$$\begin{aligned} \mathcal{D}_p = & (\mathcal{D}_{\text{train}} \setminus \mathcal{D}_{\text{sub}}) \\ & \cup \{(\mathcal{T}_\phi(x_i), y_t, r_i) : (x_i, y_i, r_i) \in \mathcal{D}_{\text{sub}}\}. \end{aligned} \quad (2)$$

The model owner trains on $\mathcal{D}_p$ and obtains f_{θ^*} . For the clean reference, we denote a model trained on the unmodified $\mathcal{D}_{\text{train}}$ under the same training protocol by $f_{\theta^{(0)}}$. At evaluation, we apply the same injection function to non-target test patches.

C. Objectives

We seek target-class predictions on triggered inputs while retaining classification accuracy on clean inputs. All expectations below denote empirical averages over test samples. Clean utility is measured by the overall accuracy of the poisoned model:

$$\text{OA}(f_{\theta^*}) = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}} [\mathbb{I}(f_{\theta^*}(x) = y)], \quad (3)$$

where $\mathbb{I}(\cdot)$ is the indicator function. Let $\mathcal{D}_{\text{test}}^{-t}$ denote the test samples with $y \neq y_t$. We measure target control by the triggered target-class rate (TCR):

$$\text{TCR} = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}^{-t}} [\mathbb{I}(f_{\theta^*}(\mathcal{T}_\phi(x)) = y_t)]. \quad (4)$$

TCR is the targeted attack-success rate on non-target test samples. We also evaluate the target-class prediction rate of the same poisoned model on the corresponding unmodified inputs:

$$\text{TCR}_{\text{clean}} = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{test}}^{-t}} [\mathbb{I}(f_{\theta^*}(x) = y_t)]. \quad (5)$$

The debiased target rate is

$$\text{DTR} = \text{TCR} - \text{TCR}_{\text{clean}}. \quad (6)$$

DTR measures the change in the target-class prediction rate after trigger injection, with the model and test samples held fixed. The separate clean reference $\text{OA}(f_{\theta^{(0)}})$ measures accuracy before poisoning.

For DSI-LAMM, we seek a support $\mathcal{B} \subseteq \{0, \dots, C-1\}$ formed by short contiguous intervals, with $|\mathcal{B}| \leq k_s$, where k_s is the spectral band budget. The motif modifies only these bands and satisfies $\|\mathcal{T}_\phi(x) - x\|_\infty \leq \epsilon$. The poisoning rate ρ controls how many training samples are modified, whereas k_s controls how many bands the trigger can affect. The classifier still receives all C bands. We report patch-level mean absolute perturbation, MAE_p , by averaging absolute changes over all P^2C entries of each evaluated patch and then over patches. Together, OA, TCR, DTR, and MAE_p describe clean utility, target control, and perturbation magnitude under the selected spectral support.

IV. METHOD

A. Overview

We propose DSI-LAMM, a directional spectral-interval motif mining framework with bounded motif generation and diversity-constrained candidate selection. Fig. 1 places these three components between the input and threat setting (1) and victim training (5). From target and background statistics, we mine a compact set of spectral intervals (2) and construct a bounded directional shift with a Gaussian spatial mask (3). We then rank non-target carriers by source-to-target prototype-margin change and post-shift target proximity, and select

them using source-class quotas and class/spatial caps (4). The selected carriers are modified and relabeled, and the victim is trained on the resulting training set. Algorithm 1 summarizes the poisoning procedure.

B. Motif Mining

We mine short contiguous intervals along the ordered spectral axis. A spectral motif is a target-discriminative interval pattern. Its support is selected from target-background statistics, not subsequence frequency. For each training sample, we extract a spectral token $z_i \in [0, 1]^C$ using band-wise medians in a centered 3×3 neighborhood. Class and block statistics are computed from tokens centered at training samples.

Let $\mathcal{I}$ be the intervals selected so far and $\mathcal{B}(\mathcal{I})$ their union. Given a band budget k_s , we consider length- L intervals that do not overlap $\mathcal{I}$ and are separated from it by at least g unselected bands. At each step, we select

$$I^* = \arg \max_{I \in \mathcal{A}(\mathcal{I})} \left\{ \frac{1}{L} \sum_{c \in I} \bar{q}_c + 0.10 \max_{c \in I} \bar{q}_c - \lambda_R R(I, \mathcal{I}) \right\}, \quad (7)$$

where $\mathcal{A}(\mathcal{I})$ contains the feasible intervals within the remaining band budget, and $\bar{q}_c$ is the smoothed band score defined below. The redundancy term $R(I, \mathcal{I})$ is the maximum absolute Pearson correlation between I and a selected interval, or zero when $\mathcal{I}$ is empty. For each training sample, the interval value is the average of its band values after each band is standardized across the training set. We use $\lambda_R = 0.15$, $L = 4$, and $g = 1$.

For each band c , we compute the Fisher discriminability between the target class and the pooled non-target background:

$$F_c = \frac{(\mu_{t,c} - \mu_{n,c})^2}{\sigma_{t,c}^2 + \sigma_{n,c}^2 + \xi}, \quad (8)$$

where $\mu_{t,c}$ and $\sigma_{t,c}^2$ are the target-token mean and variance, and $\mu_{n,c}$ and $\sigma_{n,c}^2$ are their non-target counterparts. We use $\xi = 10^{-8}$ for numerical stability.

To incorporate spatial variation, we compute the same Fisher ratio $R_{b,c}$ within each block of an 8×8 spatial partition. A block contributes when it contains at least two target and two non-target training samples. The block-wise consistency score is

$$S_c = \frac{\mathbb{E}_b[R_{b,c}]}{\text{Std}_b(R_{b,c}) + \xi}. \quad (9)$$

A large S_c combines high average discriminability with low variation across eligible blocks. When fewer than two blocks qualify, we set $S_c = 1$ for all bands, so its standardized contribution is zero.

We combine the log-transformed scores as

$$q_c = \mathcal{N}[Z(\log(1 + F)) + \lambda_S Z(\log(1 + S))]_c, \quad (10)$$

where Z standardizes across bands using the population standard deviation and denominator offset ξ , and $\mathcal{N}(v) = (v - \min v) / (\max v - \min v + \xi)$. We use $\lambda_S = 0.5$. An edge-padded moving average of width $W = 5$ produces $\bar{q}_c$.

Eq. (7) combines interval-average evidence, a local-peak term, and redundancy control. With $k_s = mL$, the selected intervals are

$$\mathcal{I} = \{I_1, \dots, I_m\}, \quad I_j = [a_j, a_j + L - 1], \quad (11)$$

with band support $\mathcal{B} = \bigcup_{j=1}^m I_j$. The mining weights are

$$w_c = \frac{\max(\bar{q}_c, 0)}{\sum_{k \in \mathcal{B}} \max(\bar{q}_k, 0)}, \quad c \in \mathcal{B}. \quad (12)$$

If the denominator is at most ξ , we use uniform weights on $\mathcal{B}$. These weights enter the carrier score.

C. Motif Generation

Given the selected band support $\mathcal{B}$, we define the injection direction by the target-background mean difference:

$$d_c = \text{sign}(\mu_{t,c} - \mu_{n,c}), \quad c \in \mathcal{B}, \quad (13)$$

with $d_c = 1$ when the difference is zero. We use a uniform per-band perturbation bound

$$\epsilon_c = \epsilon, \quad c \in \mathcal{B}, \quad (14)$$

where $\epsilon = 0.05$. Let $M \in [0, 1]^{P \times P}$ be a Gaussian spatial mask with maximum value one. It has centered 5×5 support and standard deviation $5/3$ pixels, and is zero elsewhere in the $P \times P$ patch. For spatial position (u, v) and selected band c , we inject

$$\mathcal{T}_\phi(x)_{u,v,c} = \text{clip}(x_{u,v,c} + M_{u,v}\epsilon_c d_c, 0, 1). \quad (15)$$

Bands outside $\mathcal{B}$ remain unchanged. The mask gives the largest shift at the patch center and smaller shifts toward the mask edge. The fixed direction and uniform bound specify the spectral motif; the mask specifies its spatial extent.

D. Diversity-Constrained Candidate Selection

We rank non-target samples by how the spectral shift changes their relative distances to source and target prototypes. The target anchor a uses the target-token 0.80 quantile on bands with $d_c = 1$ and the 0.20 quantile on bands with $d_c = -1$. The source reference p_i is the band-wise median of training tokens with label y_i . For a source class with fewer than five training tokens, we set $p_i = z_i$.

We measure distance on the selected support by

$$D_{\mathcal{B}}(z, p) = \frac{1}{|\mathcal{B}|} \sum_{c \in \mathcal{B}} w_c \frac{|z_c - p_c|}{\nu_c}, \quad (16)$$

where ν_c is 1.4826 times the median absolute deviation of band c over all training tokens. If this scale is below 10^{-6} , we use the band standard deviation, and then one if that value is also below 10^{-6} .

For ranking, we form a token-level proxy $\tilde{z}_i$ by setting $\tilde{z}_{i,c} = \text{clip}(z_{i,c} + \epsilon_c d_c, 0, 1)$ on $\mathcal{B}$ and leaving the remaining bands unchanged. This evaluates the spectral shift at unit mask weight; actual patch injection uses Eq. (15). For candidate x_i with source label y_i , the source-to-target prototype margin is

$$H_i(z) = D_{\mathcal{B}}(z, p_i) - D_{\mathcal{B}}(z, a). \quad (17)$$

Algorithm 1 DSI-LAMM

Require: Training set $\mathcal{D}_{\text{train}}$, target y_t , poison rate ρ , band budget k_s , interval length L , perturbation bound ϵ .

Ensure: Poisoned training set $\mathcal{D}_p$.

- 1: Extract training tokens and mine intervals $\mathcal{I}$ with support $\mathcal{B}$ using Eq. (7).
 - 2: Compute weights w_c , directions d_c , and bounds ϵ_c ; construct $\mathcal{T}_\phi$.
 - 3: Estimate target anchor a , source references p_i , and scales ν_c from training tokens.
 - 4: Compute token proxies, prototype margins, and $\text{Score}(x_i)$ for candidates in $\mathcal{C}_t$.
 - 5: Compute B , source-class quotas, and class/block caps.
 - 6: Select within-class candidates for the quotas, then fill in global score order under the caps.
 - 7: Fill any remaining budget in score order without the caps.
 - 8: Inject $\mathcal{T}_\phi$ into selected carriers and relabel them as y_t to obtain $\mathcal{D}_p$.
 - 9: **return** $\mathcal{D}_p$.
-

A negative margin places the token closer to its source reference. The margin gain is

$$G(x_i) = H_i(\tilde{z}_i) - H_i(z_i), \quad (18)$$

and the post-shift target affinity is

$$A(x_i) = -D_{\mathcal{B}}(\tilde{z}_i, a). \quad (19)$$

We rank candidates using

$$\text{Score}(x_i) = Z(G(x_i)) - \beta_c Z(H_i(z_i)) + \alpha_c Z(A(x_i)), \quad (20)$$

where Z now standardizes each term over the non-target candidate pool. We use $\beta_c = 0.5$ and $\alpha_c = 0.25$. The three terms favor a larger target-relative margin gain, a source-like initial position, and greater target proximity after the shift.

For dirty-label poisoning, the sample budget is $B = \min\{\text{round}(\rho|\mathcal{D}_{\text{train}}|), |\mathcal{C}_t|\}$. Here round rounds to the nearest integer, with ties to even. The class and block caps are $U_y = \max(1, \lceil \tau_y B \rceil)$ and $U_b = \max(1, \lceil \tau_b B \rceil)$, with $\tau_y = \tau_b = 0.25$. For n_s non-target source classes, we set a per-class quota $\ell = \min\{U_y, \max(1, \text{round}(\eta B/n_s))\}$ with $\eta = 0.5$. We first visit source classes in label order and take their highest-scoring candidates up to ℓ , subject to the block cap and total budget. We then fill the budget in global score order under both caps. If fewer than B samples have been selected, we fill remaining positions in score order without caps. The selected carriers are modified and relabeled as y_t .

Let $N_{\text{tr}} = |\mathcal{D}_{\text{train}}|$ and $J = C - L + 1$. Given the training tokens, band statistics take $O(N_{\text{tr}}C)$ time. The explicit interval correlation matrix takes $O(N_{\text{tr}}J^2)$ time and $O(J^2)$ storage, followed by $O(Jm^2)$ greedy selection. Candidate distance evaluation takes $O(|\mathcal{C}_t|k_s)$ time, followed by $O(|\mathcal{C}_t| \log |\mathcal{C}_t|)$ sorting. These computations precede victim training.

V. EXPERIMENTS

A. Experimental Setting

We evaluate DSI-LAMM on Pavia University and Salinas [23], WHU-Hi-LongKou [24], and Houston2013 [25]. Pavia University is an urban ROSIS scene, Salinas is an AVIRIS agricultural scene, and WHU-Hi-LongKou is the UAV-borne Longkou scene from the WHU-Hi collection. Houston2013 is the hyperspectral scene from the 2013 IEEE GRSS Data Fusion Contest. The four inputs contain 103, 204, 270, and 144 bands, respectively. We use hyperspectral imagery alone and min-max normalize each band over its scene to $[0, 1]$.

We form spatial class-wise splits with nominal training and validation proportions of 20% and 5%, leaving the remainder for testing. Using the block grid in Sec. IV-B, we allocate occupied blocks separately within each class to approximate these proportions. Classes occupying fewer than three blocks use pixel-wise allocation; no inter-split patch buffer is applied. We fix the resulting partitions before poisoning and reuse them across methods with seeds $\{1, 2, 3\}$. Within each seed, model initialization and training-data shuffling are reset consistently across methods and targets. One clean reference model is trained per seed and reused for all target evaluations.

All victims receive full-spectrum 11×11 patches. The compact 2-D CNN has four 3×3 convolutions with 64, 64, 128, and 128 output channels, each followed by batch normalization and ReLU. A 2×2 max-pooling layer follows the second convolution, and global average pooling, dropout of 0.1, and a linear layer produce the predictions. Training uses cross-entropy loss and AdamW for 80 epochs, with batch size 128, initial learning rate 10^{-3} , weight decay 10^{-4} , and cosine learning-rate decay. We evaluate the final epoch, without early stopping or validation-based checkpoint selection.

The main comparison uses 3% dirty-label poisoning. The DSI-LAMM band budgets are $k_s = 12, 20, 24, 16$ for Pavia, Salinas, WHU-Hi-LongKou, and Houston2013, respectively; the remaining mining and selection settings follow Sec. IV. We compare with HSI-adapted BadNets, Blended, SIG, WaNet, and WABA [9]–[13], using randomly selected non-target carriers. BadNets uses a 3×3 square with value 1. Blended mixes in the target-median spectrum, and WABA mixes its low-frequency wavelet coefficients using one-level `bior4.4`; both use mixing weight 0.2. SIG uses amplitude 0.08 and six cycles across the patch width, while WaNet uses a 4×4 control grid and strength 0.5. Every adapted trigger is clipped to the per-entry bound in Eq. (14). In the main comparison, these triggers act on all bands; the spectral budget changes trigger support, not the model input or training schedule.

We report OA for clean utility and TCR and DTR for target control. Means give equal weight to every target-seed run; TCR and DTR use non-target test samples. The patch-level mean absolute perturbation, MAE_p , averages absolute changes over all spatial and spectral entries of up to 2,000 non-target test patches per run. Table III groups the Pavia component ablations; Sec. VI-D adds fixed-carrier comparisons on Pavia and

TABLE I

MAIN COMPARISON ON FOUR HSI BENCHMARKS, AVERAGED OVER ALL TARGETS AND SEEDS 1, 2, AND 3. ALL ATTACKS SHARE THE POISONING RATE AND PER-ENTRY PERTURBATION BOUND; EXTERNAL TRIGGERS USE ALL BANDS AND DSI-LAMM USES COMPACT SUPPORT. OA AND TCR ARE PERCENTAGES, DTR IS IN PERCENTAGE POINTS, AND MAE_p IS REPORTED IN UNITS OF 10^{-3} .

Dataset	Method	OA	TCR	DTR	MAE_p
Pavia	Clean	89.4	–	–	–
	BadNets	89.4	97.7	95.4	3.72
	Blended	84.8	89.1	81.3	19.58
	SIG	89.5	100.0	98.6	38.83
	WABA	84.5	88.9	80.4	19.27
	WaNet	81.5	52.8	42.1	4.59
	DSI-LAMM	89.2	97.1	95.7	0.64
Salinas	Clean	88.4	–	–	–
	BadNets	88.5	100.0	99.1	3.72
	Blended	87.9	94.6	92.1	21.58
	SIG	88.6	100.0	99.1	38.54
	WABA	87.8	94.3	91.8	21.50
	WaNet	88.4	1.1	0.0	2.51
	DSI-LAMM	88.8	99.4	98.5	0.54
WHU-Hi-LongKou	Clean	98.1	–	–	–
	BadNets	98.0	99.7	99.4	3.72
	Blended	95.1	95.2	91.3	25.36
	SIG	98.2	100.0	99.8	36.26
	WABA	94.2	94.6	89.9	25.25
	WaNet	97.0	82.9	81.3	4.30
	DSI-LAMM	98.0	99.4	99.1	0.49
Houston2013	Clean	79.5	–	–	–
	BadNets	80.5	98.0	96.1	3.72
	Blended	77.0	75.0	65.5	9.72
	SIG	80.8	98.8	97.4	37.74
	WABA	76.9	73.6	63.8	9.50
	WaNet	77.1	23.5	11.8	2.38
	DSI-LAMM	80.6	92.2	90.4	0.60

Houston2013, including eight-band and 1% settings. Spectral Signatures and STRIP are used in the separate Pavia detection case study [15], [26].

B. Main Results

Table I shows that DSI-LAMM induces strong target behavior with a compact spectral perturbation. Its TCR ranges from 92.2% to 99.4%, while clean-test OA stays within 1.11 percentage points of the clean reference across the four scenes. The corresponding MAE_p remains below 0.64×10^{-3} . Thus, a trigger confined to a small portion of the spectrum can be learned while retaining clean classification accuracy.

BadNets and SIG achieve higher TCR on all four datasets, with larger perturbation footprints. Blended and WABA also reach high target rates on several scenes, but incur larger clean-accuracy losses than DSI-LAMM. The shared per-entry bound makes the extent and placement of the perturbation important: restricting spectral support reduces the number of modified entries without reducing the spectral information available to the victim. The next experiment holds the external trigger rule fixed and changes only this support.

TABLE II

MINED-SUPPORT TRANSFER ACROSS FOUR BENCHMARKS. A USES ALL BANDS; L RESTRICTS THE SAME ADAPTED TRIGGER TO INTERVALS MINED BY DSI-LAMM. RESULTS PAIR ALL TARGETS AND SEEDS 1, 2, AND 3. OA AND TCR ARE PERCENTAGES; Δ DTR IS L MINUS A IN PERCENTAGE POINTS. MAE RED. IS THE RELATIVE REDUCTION IN MEAN MAE_p. OA_A IS REPORTED IN TABLE I.

Dataset	Trigger	OA _L	TCR _A	TCR _L	Δ DTR	MAE red.
Pavia	BadNets	89.4	97.7	99.7	+2.7	88.3%
	Blended	88.8	89.1	98.4	+14.9	84.2%
	SIG	89.4	100.0	100.0	+0.1	88.3%
	WABA	88.7	88.9	98.2	+15.8	84.1%
	WaNet	84.7	52.8	42.7	-5.7	87.5%
Salinas	BadNets	88.7	100.0	99.8	-0.2	90.2%
	Blended	88.6	94.6	94.9	+1.7	87.4%
	SIG	88.6	100.0	100.0	0.0	90.2%
	WABA	88.8	94.3	95.1	+2.2	87.4%
	WaNet	88.5	1.1	1.1	0.0	89.8%
WHU-Hi-LongKou	BadNets	98.2	99.7	99.9	+0.3	91.1%
	Blended	98.1	95.2	99.4	+7.8	90.2%
	SIG	98.2	100.0	100.0	0.0	90.9%
	WABA	98.0	94.6	99.1	+8.9	90.2%
	WaNet	97.4	82.9	70.7	-11.6	91.4%
Houston 2013	BadNets	81.0	98.0	98.0	+0.2	88.9%
	Blended	80.4	75.0	90.2	+21.7	87.4%
	SIG	80.9	98.8	98.5	-0.3	88.9%
	WABA	80.8	73.6	89.4	+22.9	87.3%
	WaNet	79.2	23.5	8.6	-11.5	89.2%

C. Mined Support Transfer

Table II compares each adapted trigger on all bands (A) and on the target-specific intervals mined by DSI-LAMM (L). The injection rule, carrier-selection seed, perturbation bound, and victim training remain unchanged. Runs are paired by target and seed. We report clean-test OA_L alongside target control; the corresponding OA_A values appear in Table I. We average paired L-minus-A differences for Δ DTR and compute MAE reduction as $100(1 - \overline{\text{MAE}}_{p,L}/\overline{\text{MAE}}_{p,A})$, with both means taken over the paired runs.

Blended and WABA improve OA, TCR, and DTR on all four datasets after restriction to the mined support, while reducing MAE_p by approximately 84–90%. Their OA gains range from 0.76 to 4.21 percentage points; on Houston2013, both also gain more than 21 points in DTR. The gains combine a stronger trigger response with better clean utility. BadNets and SIG retain their high target rates with a fraction of the original perturbation footprint: their largest TCR decrease is below 0.3 points, and OA changes by less than 0.50 points. The selected intervals can therefore carry target associations for injection rules beyond the directional motif.

WaNet exhibits a different trade-off. Its OA increases on all four datasets, but TCR decreases on Pavia, WHU-Hi-LongKou, and Houston2013; on Salinas, DTR remains close to zero under both supports. Restricting a spatial warp to selected channels changes their alignment with the unwarped channels. These results show that reducing spectral support does not affect every injection rule in the same way. Section VI-D separately compares mined and equally compact random supports

TABLE III

ABLATION ON PAVIA UNIVERSITY WITH ALL NINE TARGETS AND SEEDS 1, 2, AND 3 (27 RUNS PER ROW). CLEAN OA IS 89.37% FOR EVERY SETTING. OA AND TCR ARE PERCENTAGES; DTR AND TCR SD ARE IN PERCENTAGE POINTS. SD IS THE POPULATION STANDARD DEVIATION OVER TARGET-SEED RUNS. ALL SETTINGS SHARE THE SPATIAL SPLIT, TRAINING SCHEDULE, SPECTRAL BUDGET, AND POISONING RATE.

Setting	OA	TCR	TCR SD	DTR
<i>Reference</i>				
DSI-LAMM	89.20	97.13	2.24	95.73
<i>Spectral support and mining</i>				
Top-scoring bands	89.15	95.87	3.16	94.40
Single sub-band	89.08	95.46	2.79	93.76
Random bands	89.25	98.59	1.80	97.22
Random intervals	88.92	97.25	1.74	95.81
Without spatial consistency	89.24	97.37	2.35	95.94
<i>Carrier selection</i>				
Top-gain, unbalanced	89.39	93.28	6.64	91.92
Without class quota	89.06	76.57	13.35	75.16
Without upper caps	89.04	97.17	2.45	95.68

with carrier identities held fixed.

D. Ablation Study

Table III compares support rules and carrier-selection settings on Pavia University. The 27 target–seed combinations share one clean reference per seed. Support controls retain the band budget, motif-generation rule, and carrier-selection procedure; the actual carriers may change because their scores depend on the support. Top-scoring bands select the largest values of q_c without interval constraints; the single sub-band maximizes the score sum in one budget-sized window. Random bands use scattered support, while random intervals match the interval count, length, and spacing rule. Each random support is sampled once per target and reused across the three training seeds. These random-support comparisons use the compact 2-D CNN. The relative performance of mined and random supports on HybridSN and CompactViT-HSI has not been evaluated.

The mined support yields higher TCR and DTR than the top-scoring and single-window controls. Random bands attain the highest target rates in this comparison, and random intervals closely match DSI-LAMM. Compact supports outside the deterministic mining rule can therefore also carry the trigger effectively on Pavia. Removing spatial consistency changes TCR by less than 0.3 percentage points, so it has little effect on aggregate target control in this case.

The carrier-selection rows compare ranking and allocation choices. Top-gain, unbalanced selection ranks carriers by directional projection gain and omits both quotas and caps. It lowers TCR and increases target–seed variability relative to the full selector. Removing only the class quota produces the largest change: TCR falls from 97.13% to 76.57%, and its standard deviation rises from 2.24 to 13.35 percentage points. Removing the upper caps while retaining the quota leaves TCR and DTR close to the full method. At this 3% Pavia operating point, removing the source-class quota changes target control far more than removing the upper caps.

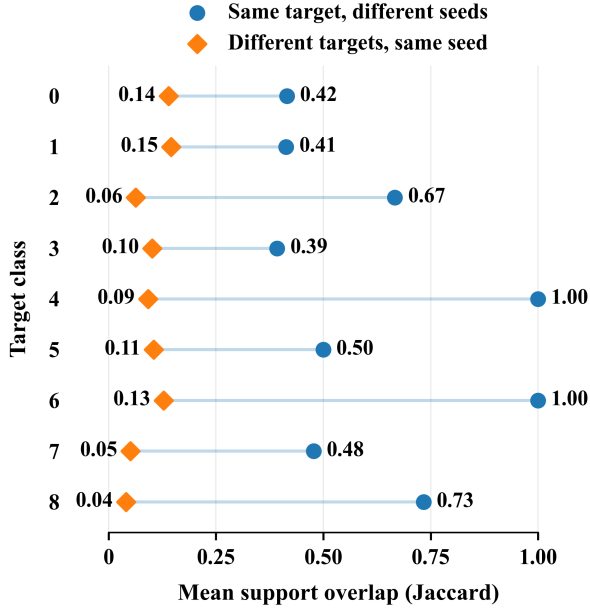


Fig. 2. Spectral-support overlap on Pavia University. Circles compare the same target across seeds; diamonds compare different targets within each seed. Each row gives the mean Jaccard values. Lines connect means, not uncertainty bounds.

VI. ANALYSIS

A. Spectral Support Structure

For two supports, Jaccard overlap is $J(\mathcal{B}_1, \mathcal{B}_2) = |\mathcal{B}_1 \cap \mathcal{B}_2| / |\mathcal{B}_1 \cup \mathcal{B}_2|$. The same-target comparison averages the three seed pairs. The cross-target comparison holds each seed fixed and compares a target with the other eight Pavia targets, giving 24 comparisons per target.

Fig. 2 gives mean overlaps of 0.622 within targets and 0.096 across targets. Targets 4 and 6 retain all 12 bands across the three splits. Other targets share support: the overlap between targets 4 and 5 averages 0.667. The layouts thus include both recurring target-dependent regions and shared regions.

Fig. 3 places these recurrences on the spectrum: target 4 retains low-index intervals, target 6 retains high-index intervals, and target 5 shares two low-index intervals across seeds while its third shifts. The budget specifies three four-band intervals; locations vary. Mean same-target overlap is 1.000 on Salinas, 0.360 on WHU-Hi-LongKou, and 0.270 on Houston2013. These within-scene comparisons show that high target control can coexist with changes in exact band locations. Sec. VI-D examines location changes with carriers fixed.

B. Victim-Model Sensitivity

We evaluate HybridSN [2] and CompactViT-HSI alongside the main 2-D CNN on Pavia. All use the 12-band directional motif, prototype-margin selection with quotas and caps, 11×11 patches, and the training schedule in Sec. V. Each victim is trained independently with seeds $\{1, 2, 3\}$ and one clean reference per seed.

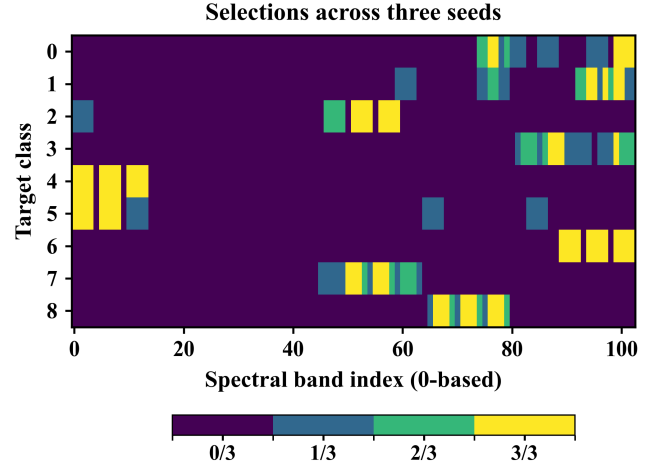


Fig. 3. Pavia band-selection counts over seeds 1, 2, and 3. Each cell records selections out of three. Band order and target indices follow the recorded supports.

TABLE IV

VICTIM-MODEL SENSITIVITY ON PAVIA UNIVERSITY, AVERAGED OVER NINE TARGETS AND SEEDS 1, 2, AND 3 UNDER THE COMMON POISONING AND TRAINING PROTOCOL. CLEAN OA AVERAGES THREE UNPOISONED MODELS. OA AND TCR ARE PERCENTAGES; DTR, Δ OA, AND STANDARD DEVIATIONS ARE IN PERCENTAGE POINTS. THE $\pm$ TERMS ARE POPULATION STANDARD DEVIATIONS OVER 27 TARGET-SEED RUNS. Δ OA USES UNROUNDED MEANS; MAE_p IS IN UNITS OF 10^{-3} .

Victim	Clean OA	DSI OA	Δ OA	TCR	DTR	MAE _p
2-D CNN	89.37	89.20	-0.16	97.13 $\pm$ 2.24	95.73 $\pm$ 2.86	0.638
HybridSN	86.93	85.95	-0.98	90.71 $\pm$ 8.68	88.33 $\pm$ 9.66	0.638
CompactViT-HSI	82.61	82.72	+0.10	87.02 $\pm$ 8.47	84.36 $\pm$ 9.66	0.638

HybridSN uses fixed PCA whitening with 15 components fitted to clean training-center spectra. Three valid 3-D convolutions have 8, 16, and 32 channels and spectral-spatial kernels $7 \times 3 \times 3$, $5 \times 3 \times 3$, and $3 \times 3 \times 3$. After merging spectral and feature-channel dimensions, a 64-channel 3×3 2-D convolution feeds dense layers of 256 and 128 units with dropout 0.4, then the classifier. Injection precedes PCA.

CompactViT-HSI maps 103-band pixels through a 1×1 convolution to 64 dimensions. Its 121 spatial tokens, a class token, and learned positions enter three pre-LayerNorm blocks with four attention heads, a 128-unit feed-forward layer, GELU, and dropout 0.1. Final LayerNorm and a linear classifier read the class token. Training is from scratch, without PCA.

In Table IV, mean clean-accuracy changes remain within one percentage point for all victims. TCR decreases from 97.13% for the 2-D CNN to 90.71% for HybridSN and 87.02% for CompactViT-HSI. The latter two show greater target-seed variation, with mean DTR above 84 points. Compact spectral triggers therefore induce associations under both convolutional and attention-based processing, with strength and variability depending on the representation.

C. Detection Case Study

Fig. 4 compares Spectral Signatures [15] and STRIP [26] on Pavia. The former scores anomalous training representations;

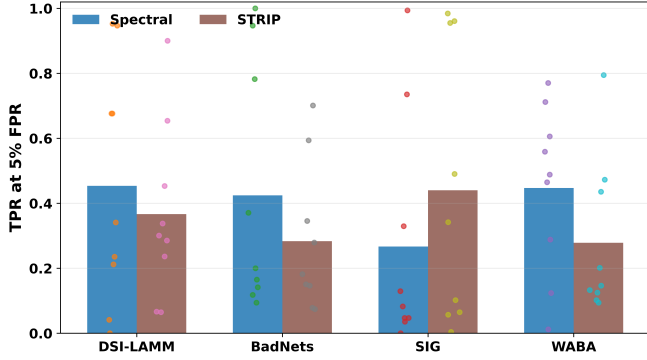


Fig. 4. Detection on Pavia University. Bars give mean TPR at 5% FPR over targets; dots give target-level values. TPR refers to poisoned training samples for Spectral Signatures and triggered test inputs for STRIP.

TABLE V

PAIRED CONTROLS WITH $\epsilon = 0.05$, ALL TARGETS, AND SEEDS 1–3. PANEL A FIXES CARRIER IDENTITIES AT 3% POISONING; M/R DENOTE MINED/RANDOM INTERVALS, AND ΔOA IS M MINUS R. PAVIA MOTIF WITH $k_s = 12$ USES FIVE SUPPORT DRAWS; THE OTHER PANEL A ROWS USE THREE. DRAWS ARE AVERAGED WITHIN TARGET–SEED PAIRS BEFORE PAIRS ARE AVERAGED. PANEL B USES MINED SUPPORT. ALL METRICS ARE IN PERCENTAGE POINTS.

A. Fixed carriers, $\rho = 3\%$			
Setting	DTR _M	DTR _R	ΔOA
Pavia, motif, $k_s = 12$	98.46	98.67	+0.04
Pavia, motif, $k_s = 8$	98.47	98.58	−0.25
Houston2013, motif, $k_s = 16$	95.63	96.27	+0.10
Pavia, Blended, $k_s = 12$	96.06	96.83	−0.61
B. Pavia, $\rho = 1\%$, $k_s = 12$			
Carrier rule	DTR ₀	DTR _p	Δ_{learn}
Prototype score (F)	0.76	92.26	91.50
Constrained random (Q)	0.76	97.75	97.00

the latter measures predictive responses to input perturbations.

At the 5% clean-sample false-alarm operating point, TPR measures the fraction of poisoned samples or triggered inputs detected. DSI-LAMM produces observable responses under both detectors: its mean Spectral Signatures TPR is close to BadNets and WABA, while its STRIP TPR lies between those two triggers and SIG. The target-level dots reveal variation around these means. A small mean input change can therefore coexist with detectable feature or prediction responses. The figure reports detection rates, not post-defense accuracy or attack removal.

D. Matched Support and Carrier Controls

We hold carrier identities fixed to compare mined supports (M) with geometry-matched random intervals (R). Constrained random selection (Q) replaces prototype scores with independent random priorities throughout the quota, cap, and budget-completion procedure. One Q set per target–seed pair is reused across supports and support draws. Direction rules, masks, perturbation bounds, initialization, and data-order seeds are shared. Blended retains its original random-carrier rule, with identities also fixed across supports. All controls use the 2-D CNN and paired clean references from these runs.

Table V(A) shows strong target control for several interval layouts on both scenes. Reducing Pavia support from 12 to eight bands preserves high DTR under both rules. Mined locations have no higher mean DTR in these fixed-carrier comparisons. Blended also works on random compact supports; its MAE_p is 3.086×10^{-3} on mined intervals and 2.225×10^{-3} on random intervals. Equal band counts and per-entry bounds thus allow different realized perturbations.

At 1% poisoning on Pavia, we fix the mined 12-band support and compare prototype ranking (F) with Q, each selecting 85 carriers. OA is 89.08% for F and 89.33% for Q. Applying the same within-model DTR calculation to the clean reference gives DTR_0 . The additional response after poisoning is $\Delta_{\text{learn}} = \text{DTR}_p - \text{DTR}_0$, where p denotes the poisoned model. Table V(B) shows a large additional response under both carrier rules; Q reaches higher DTR than F at this Pavia budget. These paired controls compare specific locations and carrier rules separately from the all-band comparisons in Table II.

E. Discussion and Limitations

DSI-LAMM constructs a training-time backdoor through an explicit spectral support. The main comparison and support-transfer experiment show that compact intervals carry target associations for several trigger forms with a small mean input change. Effective Pavia supports are not unique to the mining rule. In the scored-selector ablations at 3% poisoning on Pavia, preserving source-class coverage has a larger effect than imposing upper caps alone. Spectral layout and carrier composition therefore require joint consideration.

The threat model requires access to training spectra, locations, and editable labels. The four benchmarks test individual scenes, and overlap measures resampling within scenes. Cross-sensor, seasonal, and label-protected settings remain untested. Table IV specifies the additional victims; detector coverage is limited to two methods on Pavia.

For HSI audits, annotation provenance and class-wise inspection can help identify label corruption. Short intervals can be checked against class-conditional references, and model tests can measure prediction changes under interval perturbations. Such checks require calibration on trusted clean data and evaluation alongside clean accuracy. Developing these audits is a next step from the vulnerability studied here.

VII. CONCLUSION

We presented DSI-LAMM for studying target-related spectral vulnerabilities in HSI. It mines compact intervals, generates bounded directional motifs, and selects carriers through prototype-margin scoring and source-class coverage. Experiments on Pavia University, Salinas, WHU-Hi-LongKou, and Houston2013 show strong target control, small clean-accuracy changes, and low patch-level perturbation. The supports also carry associations for several external triggers, with repeatability varying across scenes and targets. Matched-carrier and clean-reference controls identify multiple effective layouts and

large responses acquired after poisoning. These findings motivate auditing compact spectral regions alongside annotation integrity. Future work will test interval-aware detectors and cross-scene and label-protected settings.

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